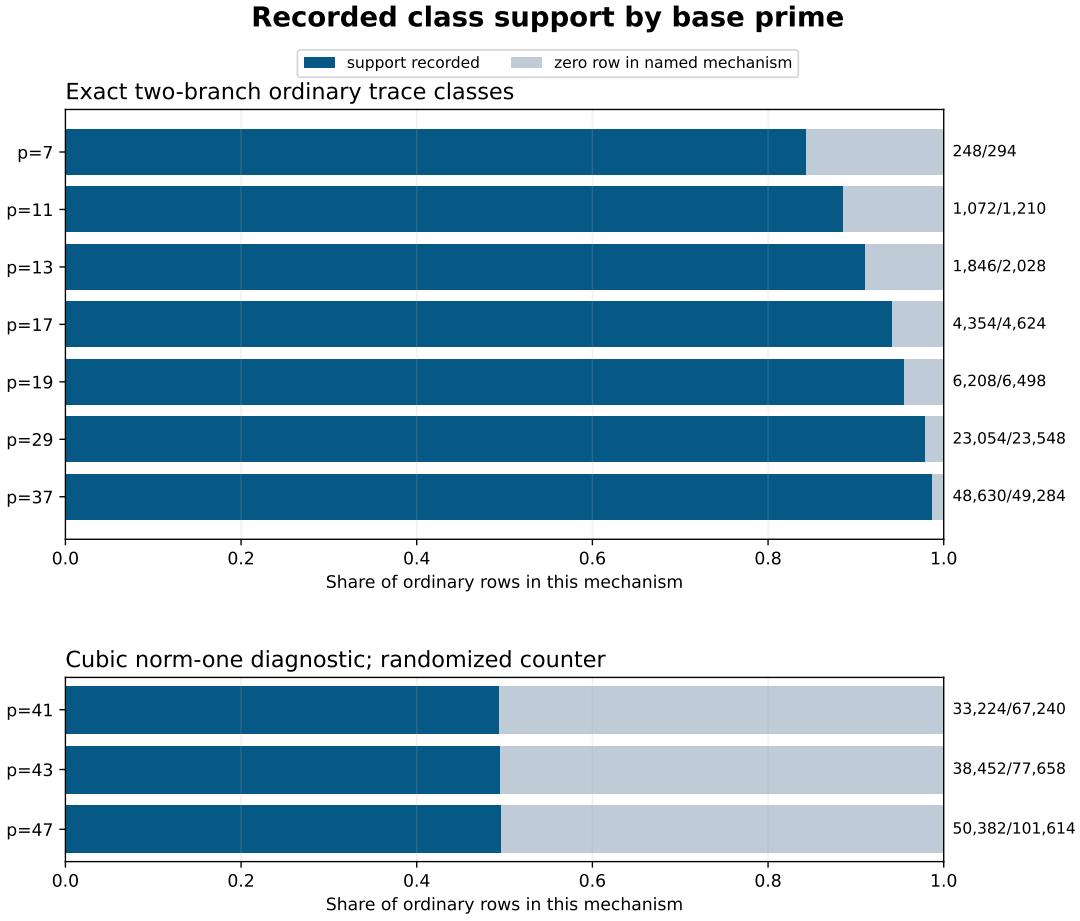
Weak-curve evidence database: exact classes, diagnostics and construction

Research report

12 October 2026 UTC

Contribution. The fixed public research corpus is now a queryable SQLite database with 339,740 class observations, 814 exact models, 21 family or condition entries with a proof obligation for each, 12 construction recipes, ten theorem records, four heuristic records and eight hypothesis entries. The cubic norm-one identity and fourth-power corollary have six checked Lean declarations. The import preserves model, class, covering and selected-subgroup evidence as separate claims.



Source: CLASS_SUMMARY.tsv. The top and bottom panels have different branch scopes and evidence grades.

The upper panel contains 87,486 complete exact ordinary rows for the two named hyperelliptic genus-three branches at $p = 7, 11, 13, 17, 19, 29, 37$: 85,412 union-positive and 2,074 union-zero. The lower panel imports 252,254 Hasse candidates at $p = 41, 43, 47$ for the earlier cubic norm-one census; 246,512 are ordinary. Those latter rows used randomized point counting, so zero labels remain diagnostic. The scopes and grades are never combined as a prevalence law.

Construction theorem and route

Let $L = \mathbf{F}_{q^3}$ with $q = p^2$ and $p > 3$. For nonzero α, put $\lambda = \alpha^q/\alpha$. The conjugate ratios telescope:

$$\lambda\lambda^q\lambda^{q^2} = \frac{\alpha^q}{\alpha}\frac{\alpha^{q^2}}{\alpha^q}\frac{\alpha^{q^3}}{\alpha^{q^2}} = 1.$$

Thus $\lambda^{q^2+q+1} = 1$. Since $q^2 + q + 1$ is odd, this parameter is a fourth power in its torus. The cubic covering route forms $E : y^2 = x(x - \alpha)(x - \alpha^q)$, then verifies the smooth genus-three cover and its map to the full Weil restriction. The prior ISO-1 theorem gives the necessary ordinary conductor screen $v_2(f_\pi) \geq 2$ for both twist signs. Exact class support uses its selected CM-order intersection or exceptional finite oracle.

The new Lean declarations prove the norm identity in a commutative multiplicative group under a cubic automorphism, and the exponent conclusion under the explicit hypothesis $\sigma(x) = x^q$. For odd p and $q = p^2$, they also construct the fourth root $\lambda^{(q^2+q+2)/4}$ from the exact torus exponent equation. They do not assume norm one. Full elliptic-curve covering, isogeny, selected-subgroup and CM bridges remain separate obligations with their exact status in `PROOF_MAP.md`. The denominator-cleared polynomial identity has native certificate `IDC1h3561e7fee541a297`.

Evidence contract

The native importer checks all 19 source hashes, full trace domains, branch-union flags, exact model identities, point-count arithmetic and 814 model/property links. Reimported table contents, TSV and SVG match. The model inventory includes 537 identities over 192–252-bit fields: 384 remain unclassified by this covering-model index and 153 are outside its stated domain. Selected-subgroup advantage is unmeasured for these 537 records. At 17 bits, 24 exact models have a positive cubic or quadratic covering-model label. The new Lean module builds and its six public roots have only propext and Quot.sound in their transitive axiom lists.

Record	Meaning and present status
Exact class row	Certified two-branch ordinary trace label on the stated finite domain.
Diagnostic row	Earlier cubic norm-one representative count; zero labels depend on randomized counting.
Model row	Exact field and Weierstrass model; an ICV1 key and source hash bind its properties.
Mechanism row	Construction and witness checklist; a selected subgroup needs its own map and cost evidence.

Next import

The published Iijima–Momose–Chao Type I/II classification is the first table-level import. Its 17 additional cases span source genera beyond elliptic curves, so equations, twists, fields and equivalence relations must be normalized before adding records. Tian's odd-prime polarized cubic-restriction construction is a separate mechanism backlog. A prospective complete two-branch census follows with frozen input and independent validation. Full source links and receipts are in `REPORT.md`, `SOURCE_AUDIT.md` and `VALIDATION.md`.